GPUOS: An Operating-System Architecture for Shared GPU Execution

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***Abstract*—Latency-sensitive inference can leave arithmetic capacity idle while occupying resources needed by other GPU work. Recovering that capacity requires a contract for interference, suspension, and resource return. Building on *The GPU Needs an Operating System*, we specify GPUOS, an architecture that represents execution as restartable tasks admitted under revocable resource leases. Its six layers connect compiler safe points, device dispatch, resource governance, interference estimates, and slack-aware admission. The design separates cooperative suspension, finite kernel slices, and driver-managed context switching, and identifies hardware services needed for enforceable pause and memory-quality guarantees. Component prototypes provide evidence on one 32-GB RTX 5090: bounded fused inference, a restricted FX-to-Triton compiler, cooperative background work, and cross-process admission. Qualified fused search reduces complete-request latency by 17.7% and 18.2% against native cached execution, while a matched CLC+PDL control remains faster. A streaming proxy admits 1.167 TFLOP/s of background work at 2.50% foreground p99 cost, but whole-step suspension recovers little work beside full-model replay. These measurements support selected mechanisms rather than a completed general-purpose implementation. The paper specifies a versioned lease API, conditional safety invariants, and checkpoint/admission feasibility conditions, with an explicit boundary between component evidence and remaining service and hardware requirements.**

I. Introduction

The useful unit of GPU ownership is rarely an entire device for the entire lifetime of an application. A model server needs a prompt response at some moments, a large KV allocation at others, and sustained memory throughput during decode. An adapter update or an evaluation job may tolerate interruption while consuming the arithmetic capacity that the server leaves available. Assigning the same indivisible device to either application hides these differences. Sharing the device without controlling interference can be worse: a throughput improvement is not useful if an interactive request misses its response-time objective.

Nourai and Bastani's *The GPU Needs an Operating System* makes this case through large-model measurements and a hardware–software research agenda [1]. We adopt its central organizing idea: GPU resources should be granted according to the urgent workload's remaining slack, rather than treating low average utilization as unconditional permission to run more work. We develop that idea into an architectural specification for GPUOS, covering process state, resource contracts, compiler checkpoints, admission, revocation, memory leases, and the hardware interface. This is a system-design paper supported by component experiments. The architecture includes services that the current prototype does not yet implement; the implementation and evaluation identify that boundary explicitly.

The resulting abstraction is a *resource lease attached to restartable work*. A lease specifies which resources a task may consume, how its progress is committed, and how much revocation delay admission must reserve. A task can run in a fused device executor, as a sequence of compiler-generated kernel slices, or in a separate protected CUDA context. These are execution backends with different costs and guarantees, not interchangeable meanings of preemption. The architecture chooses among them using a workload-specific cost model and preserves one accounting interface across the alternatives.

The prototype results expose limits as well as opportunities. Qualified fused search improves on native cached execution, but matched CLC+PDL remains faster. Cooperative sharing benefits a streaming proxy, while whole-step suspension yields little work beside full-model replay. The OS must therefore be able to reject an elaborate scheduling path when ordinary co-running is better.

The central question is whether a scheduler can rely on a pause acknowledgement without confusing preserved computation with returned resources. We address it with three contributions. First, a versioned lease contract binds executable state, storage ownership, and the resources returned at suspension. Second, a conditional safety argument and admission procedure connect those contracts to foreground protection, with explicit assumptions about enforcement and uncertainty. Third, component experiments separate useful fusion, cooperative issue suspension, finite kernel slicing, and existing context switching. The resource-vector and slack-driven motivation originates in the cited work. Our contribution is the contract and its component realization, rather than a claim of a new hardware primitive.

II. Why GPU Sharing Needs an OS Contract

A. Decode and the roofline

Consider a matrix with W weights, represented by s bytes per weight, and a decode batch of B tokens. A weight-only approximation performs $2BW$ floating-point operations while reading sW bytes. Its arithmetic intensity is therefore

$$I_{\text{weights}} = \frac{2B}{s} \quad \text{FLOP/byte.} \tag{1}$$

For two-byte weights this is B; for one-byte weights it is $2B$. KV reads, activation traffic, routing, and synchronization add costs that the expression does not capture. The device ridge $R = F_{\text{peak}}/\beta_{\text{mem}}$ compares peak arithmetic throughput

and memory bandwidth at a consistent precision. A memory-bound operation has an idealized arithmetic-utilization bound $\min(1, I/R)$. A low value is a structural property of the operation, not evidence that a dispatch queue was implemented poorly.

For a mixture-of-experts layer, the relevant batch is the batch reaching one expert. Under balanced routing, E experts and k active experts per token give an average $B_e = Bk/E$. A large global batch can consequently remain a small local matrix problem. Routing imbalance makes the average a poor description of the tail. This matters for an OS: scheduling against the global token count can overestimate the arithmetic reuse of each ready task, while the latest-finishing expert determines when dependent work is released.

a) Evidence motivating the design.: OpenInfer reports roughly 7% tensor activity for Qwen3.5-397B at a 97.2% deadline-attainment operating point on four RTX PRO 6000 cards, and 2.9% in its decode-only pool [1]. Its Kimi K2 experiments report 6–11% activity at carried loads. These are that paper's serving measurements, not our RTX 5090 observations. Its streaming control reaches approximately 99% of measured bandwidth on 24 of 188 SMs, whereas its foreground reaches only about half of peak bandwidth. This contrast motivates jointly choosing a foreground kernel and its allocation.

B. Why a scheduler alone is insufficient

Batching can increase reuse, but waiting for a larger batch consumes response time and larger steps can increase token latency. A scheduler cannot remove that tradeoff by rearranging identical non-preemptible kernels. It needs additional work that is both useful and compatible with the urgent phase. Compatibility depends on more than arithmetic intensity: a cache-resident matrix loop and an operand-streaming GEMM may issue similar tensor work while causing very different foreground interference.

Nor can the scheduler infer available SMs from low bandwidth utilization. A poorly sized foreground kernel may need many SMs to generate enough outstanding memory requests. Taking those SMs away reduces the foreground's own throughput before the background transfers a single byte. Conversely, a well-pipelined streaming kernel may saturate bandwidth with fewer workers. The resource cost of a partition therefore has two parts: the foreground's loss from a smaller execution allocation and additional interference from the admitted background workload. The architecture measures these parts separately whenever a suitable matched control exists.

C. Time, space, and authority

CUDA stream priorities influence scheduling but do not promise to interrupt already-running work [2]. Green contexts constrain spatial execution resources; MPS coordinates multiple clients; MIG provides a different isolation boundary on supported devices [3], [4]. None should be described as a software-selected, instruction-level deadline primitive merely because it changes latency in one workload. Hardware context switching, compiler checkpoints, and static partitions address different parts of the problem.

We distinguish three authorities. An application can state urgency and declare its work. A runtime can admit or withhold future slices and manage buffers it owns. A driver or device can enforce context and physical-resource boundaries. The proposed OS API exposes the strongest available guarantee without silently upgrading it. In particular, a software issue-rate limit is not a memory-controller reservation, and a CTA launch limit is not a promise that a particular set of SMs belongs exclusively to that client.

III. System Architecture and Objects

A. A six-layer decomposition

We retain the source paper's six-layer decomposition as a design vocabulary [1]. L0 provides device signals, context scheduling, queues, and enforceable resource controls. L1 defines the obligations of a shareable kernel. L2 dispatches eligible work to execution resources. L3 governs resource consumption and memory capacity. L4 estimates interference and revocation costs. L5 selects admissions within foreground slack. Table I maps this specification to the current artifact.

Figure 1 connects host submission, task descriptors, versioned code, and device execution through the contracts below.

B. Service boundary, objects, and interface

The initial service boundary is one device, one urgent class, and trusted background clients. A GPU process owns executable versions, allocations, and tasks; a task names code, tensor views, access modes, dependencies, and checkpoint state. A slice is a bounded part of a task. A lease authorizes slices to consume resources and states the suspension mode. Mutually untrusted clients require separate driver protection domains: cooperation inside a common persistent kernel is not memory or fault isolation.

The proposed API has five operations. `prepare` validates specialization, access rights, code epoch, storage, and backend capabilities without admitting execution. `grant` binds a task to a lease generation and a finite work extent. `revoke` closes further grants for that generation. `ack` publishes a committed cursor and a resource-return mask only after the corresponding backend completion condition. `retire` releases storage and code references after all consumers finish. These are logical operations; the prototypes expose parts through different interfaces.

A lease token is (p, t, e, g): process, task, executable epoch, and generation. The controller validates the entire token before accepting progress or completion. A late acknowledgement from generation g cannot close $g + 1$. An acknowledgement distinguishes stopped arithmetic, returned CTA resources, and reclaimable storage. A resident worker may provide only the first; a finite kernel returns CTA resources but can retain checkpoints. Admission must test the mask it needs, not merely whether a pause flag is set.

Layer	Architectural responsibility	Artifact boundary
L0	Context switch, bounded pause, resource enforcement	Existing driver switching is measured; no new device pause or memory-QoS interface.
L1	Restartable work and declared resource footprint	Handwritten WMMA checkpoints and strict FX/Triton slices for supported tensor graphs.
L2	Ready-work selection and dispatch	Fused per-frontier device work pool, stream-ordered worker control, and a separate host slice executor.
L3	Capacity leases, issue pacing, cache/copy governance	Bounded arenas and CTA issue pacing; no physical cache or DRAM reservation.
L4	Workload-specific interference and revoke-cost estimates	Measured configuration lookup; no predictor for unobserved workloads.
L5	Deadline-aware admission and revocation	Held-out slack selection and one-foreground IPC admission; no integrated production service controller.

TABLE I
ARCHITECTURAL SPECIFICATION AND IMPLEMENTED EVIDENCE. THE TABLE IS A CAPABILITY MAP, NOT A CLAIM THAT ALL SIX LAYERS FORM A FINISHED DEPLOYED OS.

C. Resource vectors and accounting

The proposed lease resource vector is

$$\mathbf{r} = (r_{\mathrm{SM}}, r_{\mathrm{TC}}, r_{\mathrm{L2}}, r_{\mathrm{DRAM}}, r_{\mathrm{cap}}, r_{\mathrm{link}}, r_{\mathrm{copy}}, r_{\mathrm{power}}). \quad (2)$$

These coordinates have different units: SM allocation includes registers, shared memory, and resident blocks; tensor work concerns arithmetic issue; cache/DRAM govern traffic; capacity includes reclaim cost; copies and links have directions and endpoints; power changes shared clocks. Compiler-declared logical bytes are distinct from measured physical traffic because cache reuse and competing clients change the latter. Unavailable counters remain unknown.

For foreground class f and execution state x, $C_f(\mathbf{r}, x)$ is the additional completion time caused by a grant. State includes phase, batch, context length, and device allocation. Prices need not add: cache eviction raises DRAM demand, while frequency changes affect every slice. The lookup uses measured combinations; a predictor for unseen combinations would need separate validation. Average low utilization is never sufficient admission evidence.

D. Events and executable versions

The architecture uses release/acquire publication between producers and consumers in the appropriate memory domain. A producer publishes the data and progress state before publishing completion. A consumer observes the completion event before reading those fields. A host-mapped doorbell requires system-scope ordering; a flag used only between kernels on the device can use device scope. Polling a volatile field without a publication protocol is not a replacement for these ordering rules.

Executable versions have lifetimes too. Hot-loaded code, device function pointers, and plan descriptors remain valid until every possible caller has retired. Replacing a jump-table entry does not prove that the old module is unreachable by a CTA that already loaded its pointer. The specification therefore associates a code epoch with each admitted task and reclaims an old version only after an epoch-completion condition. The existing fused executor uses precompiled handlers; general safe hot replacement remains an architectural service rather than a property established by these tests.

IV. Compiler Contract and Execution Backends

A. Safe points are semantic boundaries

A compiler cannot make an arbitrary running GPU kernel safely restartable by inserting a flag read at an arbitrary instruction. Threads may disagree about which barrier phase they are in; shared memory may hold an unpublished partial result; or an external consumer may have observed an output that the resumed task would produce again. A checkpoint is legal only when its progress and externally visible effects define an unambiguous restart state. This requirement is stronger than limiting the number of loop iterations.

For a pure pointwise tile, a commit can consist of final stores to a disjoint output region. For a matrix product, a commit may consist of a partial accumulator and the completed K extent. A reduction similarly needs its partial value and reduction interval. The next invocation consumes exactly that state. The task's output becomes visible to dependent graph nodes only after all required tiles have committed. Intermediate numerical state may remain private even when some task-local work is complete.

The proposed general contract admits opaque kernels only if their provider supplies equivalent safe-point semantics or an acceptable non-preemptible bound. Unsupported side effects, device allocation, inter-block protocols, or external collectives cannot be assumed safe merely because the front end can capture their call site. Rejecting a graph is a valid compiler result. A silent eager fallback would violate the scheduler's assumed pause contract and is therefore inappropriate for this backend.

B. Implemented FX-to-Triton lowering

The current front end captures shape-specialized, pure inference graphs with FX and also exposes a `torch.compile` backend [5]. It lowers supported operations to Triton kernels and generated pointwise expressions [6]. It accepts floating CUDA tensors with checked shape, stride, dtype, and device specializations. Supported graphs include broadcast arithmetic,

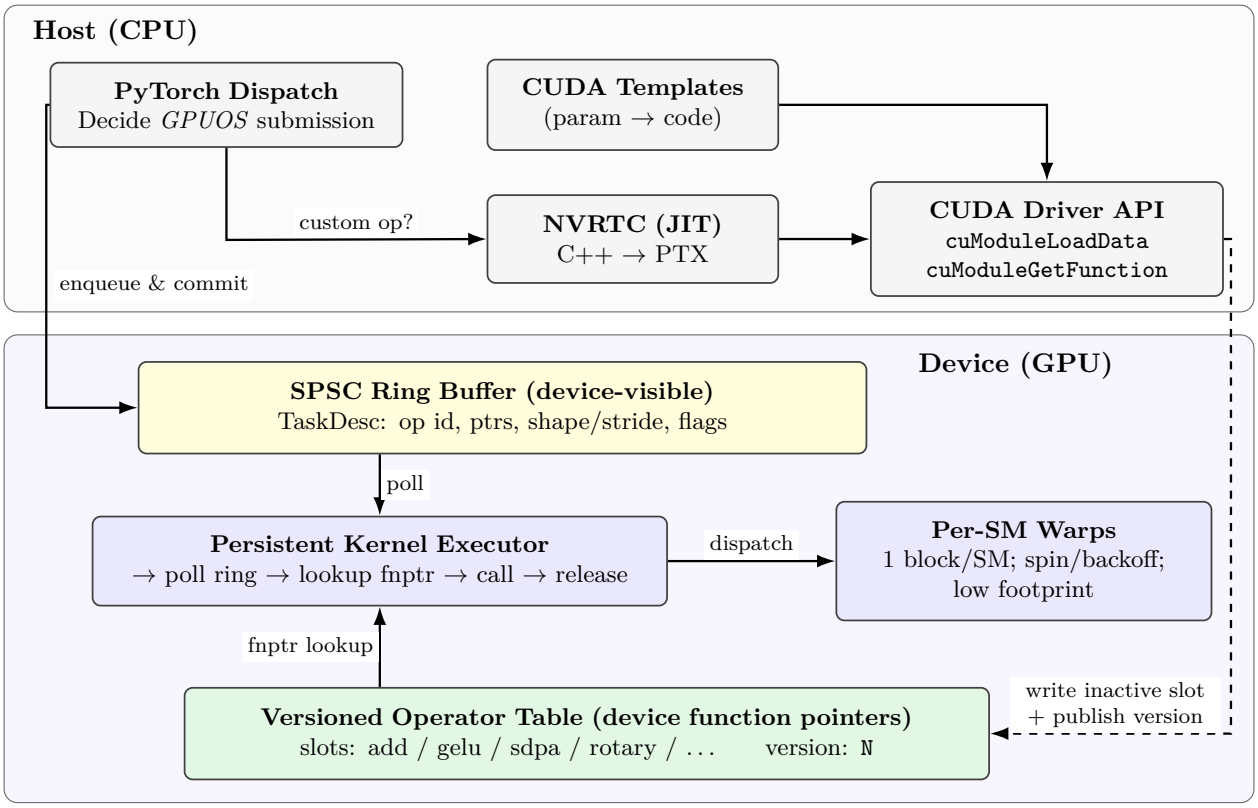


Fig. 1. **GPUOS system architecture.** Original artwork retained. The host-to-device task path, operator table, and resident executor illustrate the architectural control path. JIT publication and the pictured worker placement do not establish arbitrary hot replacement or enforced SM ownership; the capability boundary is given in Table I.

common activations, two-dimensional matrix products and linear layers, contiguous last-dimension sums and means, bounded-row softmax, and selected no-copy views. Autograd, mutation, unrecognized operations, and unsupported semantic options are rejected.

The compiler generates a sequence of stages rather than forcing every graph into a persistent megakernel. Each stage specifies its number of independent program instances, its reduction chunks, output storage, and any private partial storage. A matrix-product slice covers a bounded set of output tiles and a bounded K interval, accumulating in FP32. A sum or mean saves FP32 partials between reduction chunks. An elementwise stage maps a bounded range of logical elements through broadcasting and physical strides. A softmax row currently executes as one program and is limited to 8192 elements.

The execution cursor identifies the stage, program offset, and reduction chunk. It lives in the host executor; partial arithmetic state lives in device allocations. The executor advances through all program ranges of a chunk before advancing the chunk, allowing successive grants to change their program counts without skipping or duplicating output tiles. The same stream orders those grants. The host submission cursor can advance before device completion; it is not a committed checkpoint. An acknowledgement must wait for the stream before reporting completed progress. The prototype rejects an unordered stream change.

Every slice is a finite kernel launch. Once that launch completes, its CTA resources are returned to the CUDA scheduler. This is the important semantic difference from a resident worker that sleeps. The cost is additional launch and synchronization work, more partial-state traffic, and possibly worse cache reuse. The compiler exports resource and progress metadata so that the controller can account for these costs. A maximum program count or K extent is a bound on work geometry, not a guaranteed number of microseconds.

C. Selecting checkpoint granularity

In a serialized finite-slice approximation, let τ be useful execution time and h amortized overhead, including launch and state commit. Keeping $h/(\tau + h)$ below ϵ requires $\tau \geq h(1-\epsilon)/\epsilon$. For pause target L, publication cost p, residual checkpoint cost d, and acknowledgement cost a, feasibility requires

$$\frac{h(1-\epsilon)}{\epsilon} \leq \tau \leq L - p - d - a. \tag{3}$$

The right side assumes one in-flight slice and bounded constituent costs under admitted interference. These times differ from the experiment's K-tile count q. Empty intervals require

another backend, a relaxed objective, or refusal to share. Checkpoint size and cache effects make h and d geometry-dependent. Thus shorter slices can improve responsiveness while reducing useful throughput; a compiler must expose candidate costs, not just a maximum tile count.

D. Fused dynamic execution

The earlier fused executor, called FuseOS in the retained measurements, uses the execution and storage contract in Figure 2. It handles a different execution opportunity: changing work within a bounded model invocation. In output-driven search, scores determine the next live branch set, selected histories, and ready tiles. The model architecture alone does not fix those choices. Stable request identifiers and bounded storage plans separate changing logical ownership from physical reservations.

The evaluated Qwen3-0.6B path stores an immutable ancestor KV region and two private-history banks. Each frontier copies selected histories into the alternate bank, appends KV, and swaps banks. Precompiled handlers combine normalization and projection, gate/up and activation, and residual work. A cooperative persistent invocation executes all 28 layers and the vocabulary head for one frontier. Device work counters assign ready tiles within a phase; barriers order dependent phases. Host branch scoring remains part of the complete-request measurement. This is not a process-lifetime serving kernel with a GPU-owned external request queue.

The same arithmetic handlers support ordinary grids, programmatic dependent launch, and cluster launch control. Shape-specific graphs cache encountered widths and depths. A fixed-topology control accepts an updated device configuration without recapture. These controls matter architecturally: the OS must preserve good fused arithmetic and plan reuse even when it selects an execution backend other than persistence. It must not create an artificially weak graph baseline to justify a resident runtime.

a) A process-lifetime resident service.: The architecture also permits a resident executor to accept requests from a host-visible ring, perform sampling and stopping decisions on the device, and publish tokens through an output ring. This is the process-lifetime execution model demonstrated for Llama-3.1-8B by OpenInfer, which also assigns selected CTAs background roles at phase boundaries [1]. Such a service needs request-slot generations, backpressure, cancellation, and a rule for retiring every output consumer. A full ring must stop admissions without blocking token delivery for accepted requests. A cancelled request retires only after its current phase commits and references to its KV are released. Our frontier-level executor provides bounded model execution but does not implement this complete service loop.

E. Three meanings of suspension

GPUOS assigns an explicit suspension mode to a backend. A resident cooperative worker saves progress and stops useful issue while retaining its CTA allocation. A finite sliced backend commits and returns from the kernel, releasing CTA allocation before resumption. A separate-context backend relies on driver-managed switching, whose saved physical execution state is opaque to the application. All can preserve logical task progress, but only the second exposes an application-controlled kernel boundary with resource return.

V. Slack Admission, Dispatch, and Revocation

A. Slack is a budget with a scope

An urgent request has a release time, a completion objective, and a current estimate of its remaining service demand. The architecture distinguishes first-token latency, per-step latency, and whole-request completion: satisfying one does not prove the others. A background admission spends only slack defined at the same boundary as the controller's latency observations. Speculative tokens, variable search width, and prefill/decode transitions require the controller to change its foreground class or invalidate the corresponding calibration.

For the measured sharing experiment, the objective is a step deadline $D_f = 1.06\, Q_{.99}(T_f^{\mathrm{solo}})$. The controller reports both $Q_{.99}(T_f^{\mathrm{co}})/Q_{.99}(T_f^{\mathrm{solo}}) - 1$ and deadline attainment, the fraction of observed steps no longer than D_f. These are different statistics. Passing a p99 ratio test does not establish that every step met its deadline; observing a finite sample with no misses does not establish a deterministic deadline guarantee.

The intended admission problem is to choose background utility subject to foreground and resource constraints:

$$\max_{\mathcal{A},\mathbf{r}} U(\mathcal{A},\mathbf{r}) \quad \text{s.t.} \quad \widehat{T}_f + \widehat{C}_f(\mathcal{A},\mathbf{r}) + G_f \le D_f. \tag{4}$$

Here $\mathcal{A}$ is the admitted task set, $\widehat{C}_f$ includes predicted interference, and G_f reserves revocation and uncertainty margins. Capacity, execution-resource, and protection constraints apply independently. Utility can value completed work, throughput, or application priority. A scalar FLOP count is appropriate only when comparing work whose output and precision make that count meaningful.

The implemented policy uses a measured configuration table: select the highest-throughput numerically qualified cooperative candidate whose calibration p99 passes, freeze it, and evaluate held-out steps. It does not estimate costs for unseen combinations. The architectural controller additionally requires matching foreground phase, geometry, executable, and backend versions. A changed key invalidates the entry; a failed validation window disables later admissions until recalibration. Revocation cannot repair a missed deadline.

Quantile admission requires care. In general, adding the p99 of standalone service, interference, and revocation does not produce the p99 of their sum. If thresholds b_j have valid tail bounds $\Pr[X_j > b_j] \le \delta_j$, then $\Pr[\sum_j X_j > \sum_j b_j] \le \sum_j \delta_j$ by a union bound, without independence. Achieving a target tail budget therefore requires allocating that budget across components or directly validating their joint interval. The present paired experiment uses the latter approach; it does not establish stationarity or probabilistic guarantees for future workload changes.

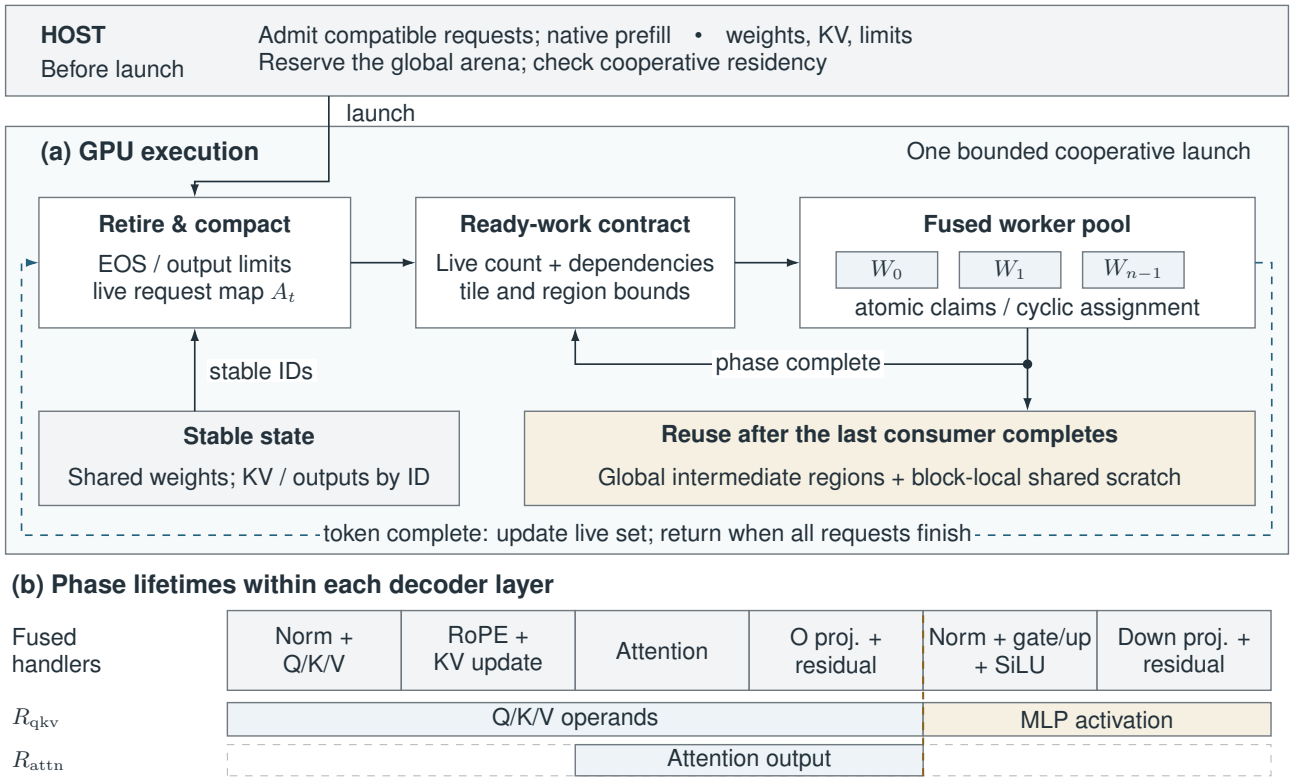


Fig. 2. **GPUOS execution and storage contract.** Original artwork retained. (a) The bounded cooperative design couples retirement, ready-tile selection, and phase completion; stable identities retain KV/output ownership. (b) Intermediate regions can be reused after their last consumer, while request-owned KV remains reserved. The token-loop sketch describes the earlier bounded decode design; the qualified search measurements in Section VII use frontier-level invocations and host branch scoring.

B. The cost model must include the foreground

A useful cost table records an allocation-only baseline in addition to a co-run result. Suppose a foreground loses a units of latency when its worker allocation shrinks, even with no background arithmetic. If a co-run then loses $a + b$, attributing all $a + b$ to memory contention misdiagnoses the problem. Conversely, assigning a separate SM subset does not remove shared cache, DRAM, power, or interconnect effects, so attributing the entire change to lost SMs is also unjustified.

C. Dispatch and safe revocation

The lease lifecycle is *prepared*, *eligible*, *running*, *revoking*, *checkpointed*, and finally *retired*. Preparation verifies tensor views, code versions, resource declarations, and storage availability. Eligibility requires predecessor completion and a valid grant. Revocation prevents new grants, requests a safe point from already-running work, and waits for the specified acknowledgement. Retirement is separate: it also requires that downstream users release the task's state and outputs.

A revocation acknowledgement means that the advertised restart state is committed and that the promised resources have been returned. It must not mean merely that one thread read an urgent flag. A resident worker can only acknowledge its weaker contract: arithmetic issue has stopped, while its CTA allocation remains. A finite-slice backend acknowledges after its CUDA stream confirms completion. The architecture attaches this semantic mode to telemetry so the policy cannot compare unlike acknowledgements as though they offered identical scheduling opportunities.

Let T_{pub}, T_{drain}, and T_{ack} denote control publication, remaining non-preemptible work, and acknowledgement cost. A software revocation estimate is

$$T_{\mathrm{revoke}} \approx T_{\mathrm{pub}} + \max_i T_{\mathrm{drain},i} + T_{\mathrm{ack}}. \tag{5}$$

It becomes a bound only if each constituent is bounded under the admitted interference and the control path is guaranteed execution capacity. A quantum defined as K tiles does not provide that guarantee by itself. Cache misses, context scheduling, and competing traffic can stretch the time of the same instruction sequence. This is why the specification retains a hardware pause requirement even when compiler slicing is available.

D. An IPC implementation of admission

The process-level prototype coordinates one urgent client and a fixed set of background clients with a shared condition variable. A background process acquires the condition lock, verifies that no urgent interval is active, and marks one slice in flight. It releases the lock, launches its finite slice, and waits for stream completion. It then clears its in-flight mark under the same lock and notifies waiters. The foreground sets urgency

under that lock and waits until every in-flight mark is clear before submitting its work.

Using one lock for the check and the in-flight transition closes an important race. If a worker first checked an event and only later published its active state, the foreground could observe the old idle state and admit itself just before that worker launched another slice. The foreground would then have a false acknowledgement. The condition-variable protocol gives admission a linearization point. It still depends on trusted workers reporting stream completion honestly; it is not a security boundary against a malicious CUDA client.

E. Admission procedure and safety conditions

The proposed control path has four steps. (1) Prepare a background task outside the urgent path and match its foreground/backend key to a qualified cost entry. (2) Reserve checkpoint storage, control-path capacity, and revocation margin; choose a candidate only if the remaining slack and resource constraints pass. (3) Under the admission lock, issue a generation-tagged finite grant. When urgency arrives, close grants before requesting revocation. (4) Resume urgent work only after acknowledgements from every conflicting lease return the required resource mask. Revalidate the key before resuming background work. The current lookup and IPC service realize different parts of this procedure; the complete cross-backend controller is specified here.

Three invariants define its safety obligations. *Commit before publish:* a cursor is visible only after all partial state it names is durable in the required memory domain. *No use after revoke:* closing generation g prevents new grants, and each acknowledgement covers all grants already issued for g. *Last consumer before reuse:* completion of a producer alone cannot reclaim its output, checkpoint, or executable epoch.

The argument is conditional on trusted clients, correct release/acquire ordering, nonoverlapping output ownership, and the backend's stated completion contract. Initially, prepared tasks publish no output and own their storage. A grant preserves ownership; a commit advances the cursor only after valid writes; dependent consumers become eligible only after publication. Closing grants and draining their recorded completions leave no authorized old-generation writer. Reference retirement then permits reuse. By induction over these transitions, pausing and resuming cannot expose an uncommitted result or reassign storage still owned by a live consumer. This is an argument about the specified protocol, not a proof of arbitrary kernels or malicious clients.

Safety alone does not imply a deadline. Foreground protection additionally requires execution capacity for the control path and upper bounds on urgent service, residual background execution, and interference whose sum fits the remaining deadline. Sleeping resident CTAs can violate the capacity premise; unbounded driver scheduling can violate the time premise. In either case, the implementation must expose a weaker service class rather than advertise a deterministic pause bound.

F. Failure, cancellation, and ownership

A general OS contract needs a terminal failure state as well as a successful checkpoint state. The specification distinguishes user cancellation, an expired admission lease, a worker that stops responding, and a device error. Cancellation or lease expiry prevents future grants; neither cancels stores or DMA operations already issued. An output is published as successful only after its completion condition holds; a cancelled partial output remains private until its in-flight accesses are known to have ended.

Recovery records bind a cursor and committed buffers to task, code epoch, and lease generation. Crash-safe replay needs versioned state: overwriting an in-place accumulator can invalidate the previous checkpoint before publication. Replay also requires retained inputs and replay-safe effects. Updating external state or entering a collective needs its own commit protocol; relaunch after timeout does not provide exactly-once effects. A timeout is missing acknowledgement, not proof that allocations are safe. A crashed worker's lease remains quarantined until the lower layer confirms completion or teardown. The IPC prototype provides neither crash recovery nor reset-surviving state.

G. Progress, fairness, and overload

Foreground protection can starve background work. A controller that pauses backgrounds for every full decode step may be safe for the foreground but economically useless when those steps occupy almost the entire timeline. The architecture therefore reports background progress alongside deadline attainment. Aging changes the order of eligible background grants but cannot create foreground slack. A promised minimum background rate requires explicit capacity reservation and foreground admission control; best-effort leases make no such promise. Sustained starvation ends a lease or triggers relocation where available, instead of retaining memory indefinitely for a task that cannot run.

VI. Memory and Resource Governance

A. Storage lifetime is independent of execution lifetime

Memory capacity often determines whether another workload can be admitted even when arithmetic is available. A process may hold model weights, request-private KV, shared prefixes, scratch, and checkpoints with very different lifetimes. The proposed allocator names these classes separately. It tracks ownership, consumers, mutability, residency, and reclaim cost, rather than treating all allocated bytes as equally revocable.

An immutable weight image can be retained while its process has no runnable task. A private KV page belongs to a logical request, not to the current position of that request in a compacted batch. A shared prefix requires reference-counted ownership and copy-on-write before divergence. Scratch can be recycled only after the last producer and consumer complete. A partial reduction is durable task state until the task retires, even though the kernel that produced it has returned. These distinctions are essential if execution is to be paused or reorganized without corrupting storage.

B. Capacity leases and eviction

The architectural capacity lease includes retained bytes, an eviction class, and an expected restoration path. Eviction can mean dropping recomputable scratch, writing checkpoints to host memory, preserving an immutable host weight image, or refusing new requests whose KV would exceed the pool. These choices have different costs and correctness requirements. A byte of unused reserved capacity can be cheap to lend; a byte holding the only copy of an active request's state may be expensive to reclaim.

Revocation of memory is a transaction. The controller first prevents new users, then waits for existing device accesses, produces a durable copy if required, updates the residency map, and finally returns the physical allocation. An application-visible pointer cannot remain valid after its backing storage has been reassigned unless the lower layer supplies the corresponding address-translation guarantee. The design therefore separates logical handles from physical mappings and requires versioned residency events for asynchronous consumers.

There is no assumption that paging is always preferable to rejection. Moving state consumes copy engines and host links, may disturb the foreground's memory traffic, and can make subsequent restoration exceed its latency objective. A controller should compare those costs with the remaining useful background work. It should also account for the cost of re-prefill if KV state is discarded. Capacity utilization without restoration cost is an incomplete admission metric.

C. Bandwidth pacing and cache pressure

The implemented resident worker can impose a minimum interval between CTA slice starts. It does not accumulate unused credits into a later burst. This provides a simple logical issue-rate control and prevents a long idle period from authorizing an unbounded foreground-interfering burst. The descriptor records operand and checkpoint bytes, but the control does not enforce a physical DRAM or L2 bandwidth ceiling.

A stronger governor would coordinate budgets across clients and resource domains. Per-worker pacing alone scales the aggregate allowed traffic with the number of workers. Reusing the same per-worker quota after doubling the grid can therefore double the intended issue envelope. The lease specifies whether a budget is per worker, per process, or per device. Global budgets require either a shared credit mechanism or hardware support; they cannot be inferred from independent local sleeps.

D. Copies, links, and power

Host transfers, peer transfers, collectives, and compute can contend for shared queues or memory bandwidth even when they use different engines. The proposed contract tags a transfer with its direction, source and destination, byte count, deadline class, and whether it is necessary to unblock urgent work. Copy admission must reserve the corresponding memory traffic as well as the copy engine. Otherwise a nominally independent DMA operation can exceed the foreground's memory budget.

Power and frequency are shared state rather than per-task constants. A background workload can alter device clocks and thereby change both foreground and background service rates. The experiments use unlocked clocks and record that limitation; they do not derive a power-aware price model from isolated timing ratios. The proposed governor would need calibrated telemetry and an explicit policy for thermal or board-power constraints. It cannot obtain a per-process power guarantee by adjusting only the number of submitted CTAs.

E. Distributed extensions and tail composition

Although the current scope is one 32-GB GPU, the architecture admits a distributed coordinator above device-local leases. A collective phase must not be admitted independently on ranks whose progress depends on one another. Pausing a rank inside an uncheckpointable collective can strand peers and invalidate a local slack estimate. Collective entry and completion therefore become explicit gang-scheduling boundaries until a lower layer provides a stronger protocol.

For illustration, if independent rank completion times have CDF $F(t)$, the latest of R ranks has CDF $F(t)^R$. A local quantile constraint is not automatically a job-level constraint. Real ranks are correlated by shared inputs, topology, and collective timing, so the independence expression is a model rather than a deployment measurement. The design requires the coordinator to price the observed job tail and to distinguish actual rank measurements from projected maxima. No multi-GPU result is claimed in this paper's single-device experiments.

VII. Prototype Evidence and Evaluation

The evaluation asks whether the architecture's component mechanisms have useful and correctly identified effects. It does not test a completed general-purpose GPU OS. We separate four evidence scopes: qualified full-model execution, resource-sharing microbenchmarks, compiler correctness, and a cross-process context-switch diagnostic. They have different timing boundaries and cannot be combined into a single system speedup.

A. Platform and qualification

The local experiments use one NVIDIA RTX 5090 with 170 SMs, 32 GB of memory, compute capability 12.0, CUDA 13.1, and driver 610.57.04. The model environment uses PyTorch 2.14.1+cu130. CPU and GPU clocks are not locked. The experiment records CPU affinity and checks competing compute processes; residual host, clock, and thermal variation remains. CUDA events measure device intervals, while host monotonic clocks measure request or process-visible latency.

The Qwen3-0.6B checkpoint is revision `c1899de`. The custom fused path must satisfy finite full-vocabulary logits, NRMSE at most 0.01, and cosine similarity at least 0.999 at every tested frontier. Final free-search paths are checked independently. A racecheck audit found a missing warp barrier

Evidence	Boundary and replication
Complete requests	Fresh prefill through final D2H/text; 3 processes, 5 requests per cell.
Search replay	Ready ancestor through fixed native search trace, with scoring timed; 3 process ratios.
Slack sharing	256 calibration and 512 held-out steps per cell, paired adjacent solo; 3 processes.
Context diagnostic	20 paired same-/different-process trials; one warmed pair of CUDA processes.
Compiler checks	Seven graph/lifecycle tests plus separate generated-kernel sanitizer checks.

TABLE II

DIFFERENT EVIDENCE SCOPES. A PROCESS RANGE IS NOT A CONFIDENCE INTERVAL, AND AN OPERATOR RESULT IS NOT A COMPLETE SERVING MEASUREMENT.

Executor	32 steps (ms)	64 steps (ms)
Native shape cache	143.3	274.9
FuseOS persistent	117.9	225.0
CLC+PDL shape cache	113.2	215.2
CLC+PDL fixed topology	121.3	232.0

TABLE III

QUALIFIED COMPLETE REQUESTS: MEDIAN OF THREE PROCESS MEDIANS, FIVE REQUESTS PER CELL. EACH INTERVAL INCLUDES FRESH PREFILL AND OUTPUT-DRIVEN SEARCH. ALL METHODS REPRODUCE THE NATIVE FINAL PATHS.

between reading and overwriting the online-softmax denominator. The retained request and replay results use the repaired, rebuilt path and fresh measurements. Earlier unqualified timing records are not substituted for the corrected evidence.

Complete-request timings include native prefill, root-state publication, all model frontiers, branch scoring, private-history copies, final decoding, and synchronization in one outer interval. Weights, tokenized inputs, workspace, and reusable graphs are resident. We do not add stage medians to construct request latency. The underlying records retain process medians and ranges. A small range is not an estimate of the probability that a future service request meets a deadline.

B. Complete-request benefit

The FuseOS executor retains an application-level benefit after the attention repair. For 32 search steps, the complete request takes 117.9 ms, compared with 143.3 ms for native shape-cached execution (1.215×). For 64 steps, the corresponding times are 225.0 ms and 274.9 ms (1.222×). The complete-request comparison includes the native GPU-length graph, which keeps graph count bounded by frontier width rather than sequence depth. Thus the comparison is not against repeated Python launches or compulsory graph reconstruction.

All 28 decoder layers and the vocabulary head execute in these intervals. The searches expand 128 and 251 branch-tokens respectively; the latter changes the live frontier between three and four branches. Both use model outputs for parent/token selection, and the corrected custom paths reproduce the final native candidate paths. The maximum request-gate NRMSE is 0.007762, below the unchanged threshold.

a) What the positive result establishes.: The complete dynamic executor is useful for this narrow-frontier workload. The result does not isolate persistence from fused arithmetic. With the same fused handlers, shape-cached CLC+PDL completes the two requests in 113.2 ms and 215.2 ms, faster than the persistent executor. The fixed-topology CLC+PDL control takes 121.3 ms and 232.0 ms. This comparison distinguishes the cost of the execution protocol from the benefit of the fused implementation. We retain the full run ranges, including prefill variation, instead of presenting the small fixed-topology differences as a universal persistent advantage.

C. Dependence on workload geometry

We next hold the search trajectory constant to remove work differences from the backend comparison. Three configurations vary ancestor length P, maximum frontier width B, and horizon H: $(128, 4, 32)$, $(512, 16, 16)$, and $(1024, 8, 32)$. Each uses a native-generated, output-dependent trace. All qualified executors perform the same selected histories and execute the selection policy during timing.

The equal-work replay shows that the narrow-frontier benefit does not extend uniformly to wider or longer-prefix cases. Persistence improves on the matched tile-grid and tile+PDL implementations, but native cached execution becomes competitive or faster as workload geometry changes. Shape-cached CLC+PDL provides another strong control. Since prefix, width, horizon, and trajectory all change between cases, this comparison defines an observed applicability boundary rather than identifying a single causal shape parameter. The maximum post-fix replay NRMSE is 0.008308; timing includes no gate-failing configuration.

D. Mechanism and larger-model controls

Matched attention controls distinguish placement and task partitioning from execution fusion. With identical 512-key jobs, replacing round-robin placement with length-aware placement gives a geometric-mean 1.234× speedup and a maximum 1.925×. Changing an already sorted ordinary grid from four partitions to finer 128-key jobs gives 1.843× geometrically, up to 3.988×. Both exploit information known before dispatch. They must not be presented as evidence that a device discovered new dynamic work more effectively.

The matched CLC phase-fusion control instead retains the same compiled entry, task geometry, resources, and global exchange buffers, switching between staged and last-producer reduction. Its geometric-mean increment is 0.994×. The change removes a launch but adds completion and ordering work. Separate cluster-transport controls likewise do not establish a large universal benefit from moving state across SMs. Their implication for the OS design is to preserve good allocation and dataflow plans before choosing a more elaborate execution mechanism.

The Qwen3.5-35B-A3B port provides another boundary. It covers routed MoE work for microbatches of one to eight

tokens and the DeltaNet decode path, while other operations remain native. Its default fused path does not beat native eager execution across the six evaluated workloads. Separating selected stages reduces two request times from 721.11 to 682.92 ms and from 554.12 to 547.39 ms. More fusion is therefore not an optimization objective by itself. This separate mixed-Q4 comparison does not establish BF16 accuracy or a port of every model operator to the new FX compiler.

E. Slack sharing and its application boundary

The resident-worker experiment runs independent $16\times16\times4096$ BF16 GEMMs with FP32 accumulators, four warps per CTA. After a configurable number q of K tiles, each warp saves its accumulator and progress. The foreground publishes an urgency flag on its own stream. Background workers observe it at checkpoints but retain their allocations while paused. Non-yielding long- and short-quantum controls isolate the effect of responding to urgency from merely changing the amount of work in a quantum.

The streaming foreground contains eight independent BF16 GEMVs with $M = 2048$ and $K = 8192$, reading 256 MiB of weights per step. Zero- and 100-μs device gaps provide distinct opportunities for background work. This is not a transformer or a Kimi shard. The model foreground executes all Qwen3-0.6B layers and the vocabulary head using native cached graphs, with a 128-token prefix, up to four branches, and a 32-step native trace. Unlike the equal-work search replay, this sharing replay excludes search scoring; unlike the complete-request experiment, it excludes prefill. That distinction is preserved in the results.

Each configuration has an adjacent standalone reference, randomized order, 256 calibration steps, and 512 held-out steps in each of three fresh processes. CUDA-event step latency includes the urgency markers and graph execution but excludes the intentional gap. Background rate counts executed K tiles over the host start-through-stop window, including startup, gaps, submission, and the last slice drain. The artifact retains 154,368 foreground timings including calibration and references.

With 100-μs gaps, the selected 64-worker, $q = 64$ configuration provides 1.167 TFLOP/s for 2.50% p99 cost. Shorter $q = 4$ checkpoints reduce cost to 0.62% but provide 0.741 TFLOP/s. The long non-yielding control produces 4.359 TFLOP/s at 23.68% cost and misses the budget. Without an intentional gap, the selected configuration provides 0.374 TFLOP/s at 3.48% cost; launch and event boundaries still exist in that case.

The table's normalized rate uses the fastest measured configuration of the same custom GEMM as its denominator. It is not a fraction of hardware peak and cannot be compared numerically with another implementation's "GPU worth" without matching its work and denominator. All nine frozen calibration selections choose the 64-worker, $q = 64$ policy, and all nine pass held-out validation. Those observations do not establish generalization to a new model, a new load trace, or a changed driver.

The full-model boundary is more important than the positive proxy point. Whole-step suspension provides only 0.039 TFLOP/s, or 0.342% of the custom solo rate. The non-yielding long-quantum control instead provides 8.779 TFLOP/s while satisfying the observed p99 budget. It lacks a prompt-yield contract, but it is a stronger performance control for that workload. The result argues for phase-sensitive admission rather than protecting an entire model step whenever a smaller interval would suffice. Negative measured latency costs are retained under unlocked clocks and are not credited as a scheduler-induced speedup.

F. Checkpoint correctness and pause observations

The handwritten worker checks all 256 outputs per warp and saved partial accumulators against independent CPU arithmetic on binary-exact operands. Resume tests include a quantum that leaves a partial GEMM. The streaming outputs remain bitwise stable and selected rows pass an independent FP32 dot-product check. The final replay's model logits match the native reference bitwise at every step. Bounded runs of the same worker entry pass memcheck, racecheck, and synccheck with kernel filters. An initial live-worker sanitizer attempt stalled and was terminated, so the bounded checks are not described as certification of all concurrent execution.

A separate device probe measures urgency publication until all 64 workers acknowledge suspension. Its p99 is 10.69, 21.85, and 738.68 μs for $q = 4$, 64, and 4096 respectively. The diagnostic polling kernel perturbs execution and excludes CPU dispatch; its timings do not enter throughput aggregates. Non-cooperative workers time out rather than acknowledging a yield. The large change with quantum illustrates why block or slice lifetime is a first-class admission cost.

The FX compiler has additional checks for broadcast/fan-out graphs, odd matrix dimensions, FP32/FP16/BF16 arithmetic, linear/GELU compositions, reductions and normalization, views, the `torch.compile` entry point, and rejection/lifecycle behavior. Tests vary the program-count grant during a partially completed reduction. Generated matrix, reduction, and pointwise kernels pass separate bounded sanitizer checks. These establish correctness for the supported cases; they do not imply coverage of arbitrary Python, arbitrary CUDA kernels, training, or the complete Qwen graph.

G. Existing hardware context switching

The context-switch diagnostic deliberately uses non-cooperative work. A background kernel launches one CTA on every SM, with more than half the per-SM shared memory reserved so that only one such CTA can reside per SM. Each CTA executes a 200-ms loop without an urgency poll. Its arithmetic state and shared-memory sentinel are verified after completion. The foreground probe reserves the same amount of shared memory, preventing it from being explained by spare same-context residency.

The device advertises compute-preemption support. GPU timestamps and SM identifiers confirm that, in every different-process trial, the probe runs while all 170 background CTA

Background policy	p50 cost (%)	p99 cost (%) [range]	TFLOP/s	Solo rate (%)	Passes
Streaming GEMV proxy, 100-μs gaps					
No yield, q4096, w64	+22.06	+23.68 [+23.64, +25.87]	4.359	38.252	0/3
No yield, q4, w64	+18.70	+20.91 [+20.77, +23.90]	2.852	24.964	0/3
Yield, q4, w64	+0.58	+0.62 [+0.58, +0.99]	0.741	6.507	3/3
Yield, q64, w64	+2.34	+2.50 [+2.32, +2.61]	1.167	10.213	3/3
Paced, q4, w32	+1.17	+1.16 [+1.15, +1.30]	0.055	0.484	3/3
Qwen3-0.6B native cached-graph decode replay					
No yield, q4096, w64	-1.80	-1.57 [-1.67, -1.46]	8.779	76.999	3/3
No yield, q4, w64	+4.44	+3.92 [+1.36, +4.24]	3.263	28.537	3/3
Yield, q4, w64	-3.40	-4.42 [-4.47, -3.14]	0.011	0.094	3/3
Yield, q64, w64	-3.25	-3.04 [-4.44, -2.48]	0.039	0.342	3/3
Paced, q4, w32	-1.58	+0.37 [-1.51, +0.42]	0.001	0.012	3/3

TABLE IV

CO-RUNNING ON RTX 5090. VALUES ARE MEDIANS OF THREE PAIRED PROCESS RATIOS; BRACKETS ARE THEIR RANGES. PASSES COUNT HELD-OUT PROCESSES WITHIN THE 6% P99 BUDGET. SOLO RATE IS NORMALIZED TO THE FASTEST MEASURED CUSTOM GEMM, NOT HARDWARE PEAK. HERE q IS K TILES PER WARP BETWEEN CHECKPOINTS AND w IS BACKGROUND CTAS, NOT RESERVED SMS.

Probe placement	p50 (ms)	p99 (ms)	Full overlap
Same context, high priority	199.910	199.930	0/20
Different CUDA process	2.000	2.046	20/20

TABLE V

WARMED SUBMISSION-TO-COMPLETION LATENCY FOR A 10-μS PROBE. FULL OVERLAP MEANS ALL 170 BACKGROUND CTA LIFETIMES ENCLOSE THE COMPLETE PROBE ACCORDING TO DEVICE TIMESTAMPS. ONE PROCESS PAIR, 20 TRIALS PER MODE.

lifetimes remain open. The corresponding register computation and shared-memory state checks pass. In the same context, higher stream priority does not interrupt the full-residency background. These observations are consistent with driver-managed hardware context preemption [7]; they do not expose its internal save format or measure the context-save operation in isolation. The reported latency includes driver scheduling and submission overhead. With only 20 trials per mode, the interpolated empirical p99 is near the maximum sample; it is not a population-tail estimate or a worst-case bound.

The process harness also runs one native Qwen foreground with one or three compiler-generated GEMM background processes on this same device. Separate contexts and a private MPS service are distinct configurations. Client PIDs and MPS server attachment are recorded, and interrupted partial background work is completed and checked after timing. The broader repeated process sweep was paused during this paper's design-focused revision; partial rounds are retained but are not used to assert an aggregate service-level speedup. Single-device multiprocessing is not presented as multi-GPU reproduction.

VIII. HARDWARE AND MODEL CO-DESIGN

The architecture requires three hardware interfaces and supporting telemetry, following Nourai and Bastani [1]. The prototype measurements motivate these requirements without implementing them.

A. S1: bounded pause with state-preserving resume

A pause primitive must identify a task/lease, preserve restart state, and acknowledge within a specified bound. Its contract states whether CTA resources return, outstanding transactions drain, and barriers remain valid. The source agenda's 5–20-μs target [1] is much stronger than our approximately 2-ms process-visible probe latency. Compiler kernel boundaries provide software control with submission and state-traffic costs; driver switching provides opaque state preservation. Neither observation supplies an application-selected microsecond deadline.

B. S2: priority and rate control in shared memory paths

SM separation alone cannot prevent L2 or DRAM interference. The desired memory interface tags transactions by task class and supports policies for urgent queueing, background rate limits, and observable enforcement. A meaningful guarantee must distinguish bandwidth from latency and must define the granularity at which priorities can change. Cache partitioning alone does not provide a bound on memory-controller queueing delay.

The controller must also avoid starving traffic that is required for forward progress. A background checkpoint or an outstanding write can be necessary before resources become reclaimable. Treating all such traffic as indefinitely deferrable can prevent the urgent lease from acquiring those resources. The proposed tagging model includes dependency-critical completion traffic and a bounded drain class in addition to foreground and background payloads. The current CTA issue pacing does not implement this memory-controller policy.

C. S3: enforceable per-kernel partitions

A partition contract must be honored across ordinary launches, recorded graphs, and model-phase changes. Static partitions remain useful; phase-adaptive sharing also needs cheap changes or queues with suitable allocations. The controller must price stream/queue transitions, synchronization, and cache disruption over a complete step. Adding each

phase's best isolated throughput can produce an infeasible schedule when every transition introduces a bubble. A software parameter alone does not establish physical partition enforcement.

D. Supporting services and observability

Priority inside an SM would complement, rather than replace, admission and memory control. Urgent instructions sharing a resident allocation can still be delayed by dependencies or transactions already issued by other work. Fast device-wide signals and queues are useful when they have explicit memory-ordering and progress guarantees. Accessible counters should separate residency, arithmetic activity, traffic, and queueing instead of exposing one ambiguous utilization number. Priority and pacing for copy engines, links, and power complete the resource-vector interface.

E. Models that produce predictable sharing opportunities

Model architecture can make the scheduling problem easier without replacing the OS contract. Balanced expert routing and similar expert sizes can reduce phase variability. Linear-attention components can avoid some context-length growth in the recurrent state update. Speculative or multi-token processing can change the arithmetic reuse of a step. Each choice also changes numerical semantics, latency, or memory use and must be evaluated with the actual model. We do not infer a service speedup from regularity alone.

IX. Related Work and Scope

The resource-vector and slack-budget framing is inherited from *The GPU Needs an Operating System* [1]. We turn that agenda into a contract joining checkpoint state, resource return, and storage/code lifetime. The distinctive question is what admission may conclude from an acknowledgement, including cases where useful issue stops but the allocation remains occupied.

Prior systems already schedule and preempt GPU work. REEF uses reset-based preemption for suitable DNN inference kernels and dynamically pads urgent kernels with background work [8]. Its replay assumptions differ from saving a reduction cursor and accumulator, and its results preclude a blanket claim that microsecond preemption is absent from existing research. XSched abstracts preemptible command queues across accelerator capabilities [9]. Our lease is complementary: it describes retained state, resource-return semantics, and the interference budget surrounding a dispatch mechanism. We do not claim broader portability or faster preemption than either system; neither is an experimental baseline in the local artifact.

Salus couples iteration scheduling with memory sharing in a consolidated execution service [10]. Gandiva uses repetitive training behavior to guide time slicing and placement across a cluster [11]. These establish sharing and workload predictability as prior ideas. GPUOS focuses on the contract between an inference deadline, a revocable slice, and state whose lifetime exceeds that slice. Device-local leases do not replace a cluster scheduler or imply tested migration.

FX and Triton provide capture and tiled code generation [5], [6]; our backend adds a restricted checkpointed plan. MPK compiles tensor programs into megakernel execution [12]. Meld combines dynamic work placement and cross-SM cooperation with just-in-time reduction [13]. Fusion changes the work and synchronization that a lease contains, whereas the lease governs when that work may run and what must survive suspension. The matched controls separate those effects.

CUDA stream priorities, green contexts, MPS, and context switching provide distinct lower-layer mechanisms [2]–[4], [7]. A lease selects a mechanism whose contract matches its needs. PagedAttention addresses KV allocation and sharing [14]; compute revocation additionally requires that outstanding readers and writers retire before those pages are reclaimed. These relationships motivate an interface between scheduling and memory management rather than a replacement for each underlying system.

X. Limitations and Research Questions

The complete object model remains an architectural specification. The current artifact contains bounded fused inference, a restricted graph compiler, cooperative workers, an IPC admission prototype, and diagnostics. It lacks an integrated multi-tenant service, arbitrary-kernel transformation, a learned interference predictor, a general memory-lease manager across protected contexts, and the proposed hardware QoS controls. Correctness of supported graphs and component numerical checks is not a proof of whole-system isolation, fault recovery, or deadline satisfaction.

The measurements use one consumer Blackwell GPU with unlocked clocks. They neither reproduce the source paper's RTX PRO 6000 model serving nor establish transfer to AMD, TPU, SRAM-centric accelerators, or multi-node deployments. Roofline reasoning can motivate those investigations, but partition granularity, memory-system behavior, driver scheduling, and communication must be remeasured on each target. In particular, the observed context-switch diagnostic cannot specify a worst-case scheduling bound.

XI. Conclusion

GPUOS specifies what a shared GPU task may consume, what suspension preserves, and which resources an acknowledgement returns. Versioned leases connect compiler checkpoints, storage lifetime, and slack admission. The conditional safety argument identifies the assumptions needed for that contract; component evidence demonstrates selected opportunities and exposes backend-specific costs. A general deployed OS still requires the integration and enforcement services identified in this design.

a) Artifact and drafting disclosure.: The repository includes raw timings, hashes, numerical checks, and build instructions. Both figures preserve original design artwork; incomplete sweep rounds are excluded from aggregate claims. AI-assisted tools supported code, analysis, and prose. Results derive from retained measurement records.

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